When Reading becomes Boring, BBQ makes it Caring!

AMAZING MIDDLE SECTION

SAQ

SECTION-C

When Exam comes near, BBQ removes Fear!

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SAQ CHAPTERS

32 MARKS

★★★	1.	ELECTRIC CHARGES AND FIELDS (4 Marks)	32 - 34
★★★	2.	ELECTROSTATIC POTENTIAL (4+4 Marks)	35- 39
★★★	3.	CURRENT ELECTRICITY (4 Marks)	40 - 42
★★★	4.	MAGNETISM AND MATTER (4 Marks)	43 - 44
★★½	5.	ELECTROMAGNETIC INDUCTION (4 Marks)	45 - 46
★★	6.	ALTERNATING CURRENT (4 Marks)	47 - 48
★★½	7.	WAVE OPTICS (4 Marks)	49 - 50
★★★	8.	DUAL NATURE (4+4 Marks)	51- 54
★★★	9.	ATOMS (4 Marks)	55 - 57
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Super Scoring Pages

★★★SSP★★★

Page 32, 33, 35, 36,37, 40, 43, 45, 46, 48, 49, 51, 52, 54, 55,59

DEAR DUDE!

Please note that SAQs in each chapter are Jumbled, and

Easy SAQs are brought to the beginning.

1 ELECTRIC CHARGES & FIELDS

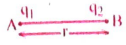
4 Marks

14. State and explain Coulomb's inverse square law in electricity. [AP 17,18,23]

A: **1) Coulomb's inverse square law:**

The force of attraction or repulsion between two electric charges is directly proportional to product of their charges and is inversely proportional to the square of distance between them and acts along the line joining the charges.

2) Explanation: Consider two point charges q_1, q_2 separated by a distance r.



3) From Coulomb's law: i) $F \propto q_1q_2$ ii) $F \propto \frac{1}{r^2}$

$$\therefore F \propto \frac{q_1q_2}{r^2} \Rightarrow F = k\frac{q_1q_2}{r^2} \Rightarrow F = \frac{1}{4\pi\varepsilon_0}\frac{q_1q_2}{r^2}$$

Here ε_0 is the permittivity of free space and $\frac{1}{4\pi\varepsilon_0} = 9 \times 10^9\ Nm^2C^{-2}$

4) If the charges are in a medium with permittivity ε then $F_{med} = \frac{1}{4\pi\varepsilon}\frac{q_1q_2}{r^2}$

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15. Define intensity of electric field at a point. Derive an expression for the intensity due to a point charge. [AP 16]

A: **1) Intensity of Electric Field (E):**

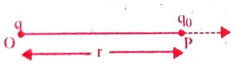
The intensity of electric field at a point in space is defined as the force (F) experienced by a unit positive test charge (q) placed at that point in the field.

2) Formula: $E = \frac{F}{q_0}$ **SI Unit:** N/C

3) Intensity due to a Point Charge :

Consider a point charge q kept at point O.

Let a test charge q_0 be kept at P, which is at a distance r from O.



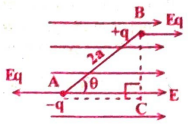
4) The force between the two charges is $F = \frac{1}{4\pi\varepsilon_0}\frac{qq_0}{r^2} \Rightarrow \frac{F}{q_0} = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}$

$$\therefore E = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\left(\because E = \frac{F}{q_0}\right)$$

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16. Derive an equation for the couple acting on an electric dipole in a uniform electric field. [AP 16,25]

A: 1) Consider an electric dipole consisting of two equal and opposite charges +q & −q separated by a distance 2a.



2) Let the dipole be placed in a uniform electric field of intensity E at an angle θ to the direction of electric field.

3) The electrostatic forces acting on the charges q and −q are Eq and −Eq respectively. These two forces form a couple.

4) Couple τ = Force × Perpendicular distance between the forces

$\tau = Eq \times BC$.....(1)

5) From Δ ABC, $\sin\theta = \frac{BC}{AB} = \frac{BC}{2a} \Rightarrow BC = 2a\sin\theta$.....(2)

From (1) and (2), $\tau = Eq(2a\sin\theta) = (2aEq)\sin\theta = q(2a)E\sin\theta$

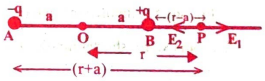
6) But we know dipole moment $p = q(2a)$

$\therefore \tau = pE\sin\theta$.....(3)

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17. Derive an expression for the intensity of the electric field at a point on the axial line of a dipole. [BMP][AP 18,19,22]

A: 1) Consider an electric dipole AB consisting of two equal and opposite charges −q & q separated by a distance 2a.



2) Let P be a point on the axial line of the dipole at a distance r from its centre O.

Then distance PB = (r − a) and PA = (r+a).

3) Intensity of electric field at P due to +q is $E_1 = \frac{1}{4\pi\varepsilon_0}\frac{q}{(r-a)^2}$.....(1)

4) Intensity of electric field at P due to −q is $E_2 = \frac{1}{4\pi\varepsilon_0}\frac{q}{(r+a)^2}$.....(2)

5) The resultant intensity of electric field due to the dipole at point P is $E = E_1 - E_2$

$$\Rightarrow E = \frac{1}{4\pi\varepsilon_0}\left(\frac{q}{(r-a)^2} - \frac{q}{(r+a)^2}\right) = \frac{q}{4\pi\varepsilon_0}\left[\frac{(r+a)^2-(r-a)^2}{(r^2-a^2)^2}\right]$$

$$= \frac{q}{4\pi\varepsilon_0}\left[\frac{4ra}{(r^2-a^2)^2}\right] = \frac{q}{4\pi\varepsilon_0}\left[\frac{2\times 2ar}{(r^2-a^2)^2}\right] = \frac{1}{4\pi\varepsilon_0}\left[\frac{2(q2a)r}{(r^2-a^2)^2}\right]\left[\because (a+b)^2-(a-b)^2 = 4ab\right]$$

$$\therefore E = \frac{1}{4\pi\varepsilon_0}\left[\frac{2pr}{(r^2-a^2)^2}\right] \quad .....(3)\ [\because p = q(2a)]$$

6) For shorter dipole we have (a<<<r) so that a^2 can be neglected. $\therefore E = \frac{1}{4\pi\varepsilon_0}\frac{2pr}{r^4} = \frac{1}{4\pi\varepsilon_0}\frac{2p}{r^3}$

Thus, the intensity of electric field on the axial line is $E = \frac{1}{4\pi\varepsilon_0}\frac{2p}{r^3}$

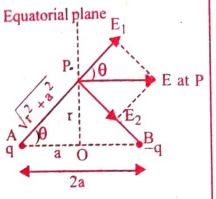
18. Derive an expression for the intensity of the electric field at a point on the equatorial plane of an electric dipole. [AP 17,19,24]

A : 1) Consider an electric dipole AB consisting of two equal and opposite charges $-q$ & q separated by a distance **2a**.
Let P be a point on the equatorial plane of the dipole at a distance r from its centre.

2) Electric field at P due to $+q$, $-q$ is $E_1 = \frac{1}{4\pi\varepsilon_0}\left(\frac{q}{r^2+a^2}\right)$.....(1); $(\because AP^2 = r^2 + a^2)$

3) Electric field at P due to $-q$ is $E_2 = \frac{1}{4\pi\varepsilon_0}\left(\frac{q}{r^2+a^2}\right)$.....(2) $(\because BP^2 = r^2 + a^2)$

From figure, it is clear that the y-components of E_1 and E_2 are equal and opposite. Hence they cancel each other.
But the sum of x-components gives the resultant field E at P.



4) Magnitude of resultant electric field is $E = E_1\cos\theta + E_2\cos\theta$

$= \frac{1}{4\pi\varepsilon_0}\left(\frac{q}{r^2+a^2}\right)\cos\theta + \frac{1}{4\pi\varepsilon_0}\left(\frac{q}{r^2+a^2}\right)\cos\theta = 2\times\frac{1}{4\pi\varepsilon_0}\left(\frac{q}{r^2+a^2}\right)\cos\theta$

5) From Δ PAO, $\cos\theta = \frac{AO}{AP} = \frac{a}{\sqrt{r^2+a^2}}$

$\therefore E = 2\frac{1}{4\pi\varepsilon_0}\left(\frac{q}{r^2+a^2}\right)\frac{a}{\sqrt{r^2+a^2}} = \frac{2qa}{4\pi\varepsilon_0(r^2+a^2)^{3/2}}$

6) But we know dipole moment $p = q(2a)$ $\therefore E = \frac{1}{4\pi\varepsilon_0}\frac{p}{(r^2+a^2)^{3/2}}$.....(3)
For shorter dipole we have $(a<<r)$ so that a^2 can be neglected.

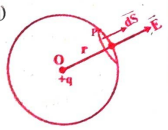
7) From (3), for shorter dipole, $E = \frac{1}{4\pi\varepsilon_0}\frac{p}{r^3}$

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19. State Gauss's law in electrostatics and prove it. [AP, 20]

A: 1) **Gauss's Law:** The total electric flux (ϕ) through any closed surface is equal to $1/\varepsilon_0$ times the net charge (q) enclosed by the closed surface. Thus $\phi = \frac{1}{\varepsilon_0}(q)$

2) Proof of Gauss's Law
Suppose an isolated positive point charge q is situated at the centre O of a sphere of radius r.
Consider a point P on the sphere enclosed in a small closed surface area dS.



3) The electric field intensity at P is $E = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}$.....(1)

4) Total electric flux over the entire surface of sphere is $\phi_E = \oint_S \vec{E}\cdot\vec{dS} = \oint_S E\,dS\cos\theta$
Here, $\vec{E}$ and $\vec{dS}$ are parallel. Hence $\theta = 0^\circ \Rightarrow \cos\theta = \cos 0^\circ = 1$

5) $\therefore \phi_E = \oint E\,dS = E\oint dS = E[S] = E\times 4\pi r^2$ .....(2) [$\because$ Surface area of sphere $S = 4\pi r^2$]

6) From (1) & (2), $\phi_E = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\times 4\pi r^2 = \frac{q}{\varepsilon_0} = \frac{1}{\varepsilon_0}(q)$

$\therefore \phi = \frac{1}{\varepsilon_0}(q)$ Hence Gauss law is proved.

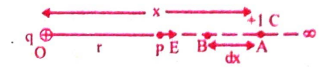
2 ELECTROSTATIC POTENTIAL

4 M+ 4M = 8 Marks

20. Derive an expression for the electric potential due to a point charge. [AP 19,23,25]

A : Potential due to a Point Charge:

1) Consider a positive point charge q at O.
Let P be a point at a distance r from O.



2) Consider a unit positive test charge (+1 C) at A on the line OP at a distance x from O.

3) At A, electrostatic force on unit positive charge (+1 C) is $F = \frac{q\times 1}{4\pi\varepsilon_0 x^2} = \frac{1}{4\pi\varepsilon_0}\frac{q}{x^2}$ ....(1)

4) If dW is the small work done in moving the unit positive charge at A against the force through a small distance (AB) = dx, then $dW = -F\,dx$ [$\because$ the two positive charges repel as $\theta = 180^\circ$]

5) From (1), $dW = -\frac{1}{4\pi\varepsilon_0}\frac{q}{x^2}dx$

6) Total work done(W) is obtained by integrating this between the limits from ∞ to r.

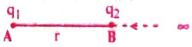
$$W = -\int_\infty^r \frac{1}{4\pi\varepsilon_0}\frac{q}{x^2} = \frac{-q}{4\pi\varepsilon_0}\int_\infty^r \frac{1}{x^2}dx = \frac{q}{4\pi\varepsilon_0}\left[\frac{1}{x}\right]_\infty^r = \frac{q}{4\pi\varepsilon_0}\left(\frac{1}{r}\right)\left(\because \int\frac{1}{x^2}dx = \frac{-1}{x}\right)$$

The electric potential due to a point charge is this work done. $\therefore V = \frac{1}{4\pi\varepsilon_0}\left(\frac{q}{r}\right)$

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21. Derive an expression for the electrostatic potential energy of a system of two point charges and find its relation with electric potential of a charge.

A: Potential Energy of System of Two Charges :



1) Consider a positive charge q_1 located at A.
Let another charge q_2 be brought from infinity to the point B.
Let r be the distance between A and B.

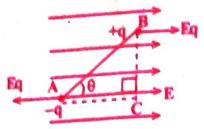
2) Potential at B due to charge q_1 is $V_1 = \frac{1}{4\pi\varepsilon_0}\frac{q_1}{r}$

3) The work done to bring the charge q_2 from infinity to B is $W = V_1(q_2) = \frac{1}{4\pi\varepsilon_0}\frac{q_1q_2}{r}$
As this is the total work done to build the system, it gives the potential energy of the system.

4) Potential energy $U = \frac{1}{4\pi\varepsilon_0}\frac{q_1q_2}{r}$
The same result is obtained while bringing the charge q_1 first and q_2 later.

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22. Derive an expression for the potential energy of an electric dipole placed in a uniform electric field.



A: 1) Consider an electric dipole AB consisting of two equal and opposite charges –q & q separated by a distance **2a** placed in a uniform electric field E.

2) The torque on the dipole is $\vec{\tau} = \vec{p} \times \vec{E} = pE \sin\theta$

3) If an external torque τ_{ext} is applied to rotate the dipole from θ_0 to θ_1, the work done is given by $W = \int_{\theta_0}^{\theta_1} \tau_{ext}\, d\theta = \int_{\theta_0}^{\theta_1} pE \sin\theta\, d\theta = pE[-\cos\theta]_{\theta_0}^{\theta_1} = pE(\cos\theta_0 - \cos\theta_1)$

4) This work done is stored as the potential energy in the system.

If $\theta_0 = \pi/2$ and $\theta_1 = \theta$, the potential energy of the dipole at an angle θ, in uniform electric field is $U = pE(\cos\pi/2 - \cos\theta) = pE(0 - \cos\theta) = -pE\cos\theta = -\vec{p}.\vec{E}$

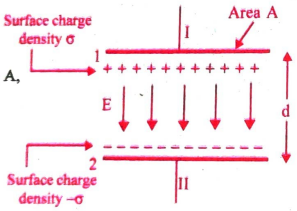
$\therefore U = -\vec{p}.\vec{E}$

This is the expression for potential energy of dipole at an angle θ in a uniform electric field.

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23. Derive an expression for the capacitance of a parallel plate capacitor. **[BMP][AP 16,18,22,24]**

A: Capacitance of a parallel plate capacitor :



1) Consider a parallel plate capacitor consisting of two parallel plates of area A, separated by a small distance 'd'.

2) Let V be the potential difference between two plates.

3) The charges of the plates are Q and –Q.

4) The plate 1 has uniform surface charge density σ and the plate 2 has uniform surface charge density –σ.

5) The electric field due to plate 1 is $\frac{\sigma}{2\varepsilon_0}$. The electric field due to plate 2 is $-\frac{\sigma}{2\varepsilon_0}$.

6) Electric field between two charged plates is $E = \frac{\sigma}{2\varepsilon_0} + \frac{\sigma}{2\varepsilon_0} = 2\left(\frac{\sigma}{2\varepsilon_0}\right) = \frac{\sigma}{\varepsilon_0}$

We know $\sigma = Q/A$

$\therefore$ The field inside the capacitor is $E = \frac{Q}{A\varepsilon_0}$ .....(1)

7) But, $E = \frac{V}{d}$ and Q=CV From (1), $\frac{V}{d} = \frac{CV}{A\varepsilon_0} \Rightarrow C = \frac{\varepsilon_0 A}{d}$

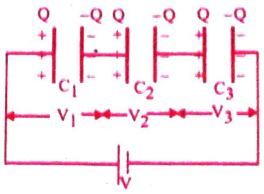
$\therefore$ Capacitance of the parallel plate capacitor is given by $C = \frac{\varepsilon_0 A}{d}$

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24. Derive an expression for the equivalent capacitance of a combination when three capacitors are connected in series.

A: **1) Series Combination:** In series combination, the second plate of first capacitor is connected to first plate of second capacitor and so on. Also, first plate of first capacitor and second plate of last capacitor are connected to opposite terminals of a battery.

2) Equivalent capacitance in series:



Let us suppose that three capacitors of capacitances C_1, C_2, C_3 are connected in series to a potential difference V.

Let Q be charge in each capacitor which remains constant in each capacitor.

3) Let V_1, V_2, V_3 be the potential differences between the plates of the capacitors C_1, C_2, C_3 and C be the equivalent capacitance

4) Total potential differences is $V = V_1 + V_2 + V_3$ ...(1)

5) $\therefore V = \frac{Q}{C_1} + \frac{Q}{C_2} + \frac{Q}{C_3}$ $\left[\because V_1 = \frac{Q}{C_1}, V_2 = \frac{Q}{C_2} \text{ and } V_3 = \frac{Q}{C_3}\right]$

6) $\frac{V}{Q} = \left[\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}\right] \Rightarrow \frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$ .....(2)

7) For n number of capacitors in series, $\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \text{.....} + \frac{1}{C_n}$

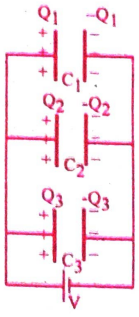
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25. Derive an expression for the equivalent capacitance of a combination when three capacitors are connected in parallel. **[AP 16]**

A: Parallel Combination:

1) In parallel combination, the first plates of all the capacitors are connected to a common terminal and second plates of all the capacitors are connected to common terminal and these terminals are connected to opposite terminals of the battery.

2) Equivalent capacitance in parallel:



Consider three capacitors of capacitances C_1, C_2, C_3 connected in parallel to a potential difference V.

In parallel combination of capacitors, potential difference V remains the same on all the capacitors.

3) Let Q_1, Q_2, Q_3 be the charges stored on the plates of capacitors C_1, C_2, C_3. In parallel combination, the total charge(Q) is equal to sum of charges stored in the capacitors.

4) $\therefore Q = Q_1 + Q_2 + Q_3$ .....(1)

$Q = C_1V + C_2V + C_3V$ $[\because Q_1 = C_1V, Q_2 = C_2V, Q_3 = C_3V,]$

5) $\frac{Q}{V} = C_1 + C_2 + C_3 \Rightarrow C = C_1 + C_2 + C_3$ .....(2)

This is the equivalent capacitance of a combination.

6) For n number of capacitors in parallel, $C = C_1 + C_2 + C_3 \text{.....} + C_n$

26. Derive an expression for the energy stored in a capacitor.

A: Energy Stored in a Capacitor :

1) When a capacitor of capacitance C is connected to a battery of potential difference V, the capacitor gets charged to Q. The work done to move the charge is stored in the capacitor in the form of potential energy called energy stored (U).

2) If dW is the small work done to store a charge of dQ at the potential drop V, we have $dW = V\,dQ$.....(1)

3) The total work done to store the charge can be obtained by integrating it in the limits from 0 to Q.

$$W = \int_0^Q V dQ \quad \text{But we know } C = \frac{Q}{V} \Rightarrow V = \frac{Q}{C}$$

4) $\therefore W = \int_0^Q \frac{Q}{C} dQ = \frac{1}{C}\int_0^Q Q dQ = \frac{1}{C}\left(\frac{Q^2}{2}\right) = \frac{Q^2}{2C}$.....(2) $\left(\because \int x dx = \frac{x^2}{2}\right)$

5) This work done is stored in the form of energy U.

$$\therefore U = \frac{Q^2}{2C} = \frac{(CV)^2}{2C} = \frac{CV^2}{2} \quad \left(\because \frac{Q}{C} = V \Rightarrow Q = CV\right)$$

$\therefore$ Energy Stored in a Capacitor is $U = \frac{1}{2}CV^2$

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27. Explain the behaviour of dielectrics in an external field. [AP 19]

A : 1) Dielectrics are non-conducting substances.

The molecules of a dielectric may be polar or non-polar.

2) Polarization: When a dielectric with non-polar molecules is placed in an external electric field, the positive and negative charges of non-polar molecules are displaced in opposite directions. Thus induced dipole moments are developed in the dielectric by the external field. Hence the dielectric is said to be polarised by the external field. All the induced dipole moments of the non-polar molecules add up to give a net dipole moment to the dielectric.

3) In dielectric with polar molecules, in the absence of external electric field, the different permanent dipoles are oriented randomly due to **thermal agitation.** Hence, the total dipole moment becomes zero. When an external electrical field is applied, the individual dipole moments tend to align with the field. As a result, a net dipole moment develops in the dielectric in the direction of external field. Hence the dielectric gets polarised.

4) Thus in either case, whether polar or non-polar, a dielectric develops a net dipole moment in the presence of an external field. The dipole moment per unit volume is called polarisation (P).

5) For linear isotropic dielectrics, $P = \chi_e E$

Here χ_e is electric susceptibility and E is intensity of external field.

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PROBLEMS

28. Three capacitors of capacitances 2pF, 3pF and 4pF are connected in parallel.
a) What is the total capacitance of the combination?
b) Determine the charge on each capacitor if the combination is connected to a 100V supply. **[BMP] [AP 17] [TS 15]**

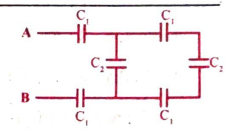
Sol: a) $C_p = 2+3+4 = 9pF$

b) For each capacitor, V = 100 Volt

$q_1 = C_1V = 2\times100 = 200pC$; $q_2 = C_2V = 3\times100 = 300pC$; $q_3 = C_3V = 4\times100 = 400pC$

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29. If C_1 = 3 pF and C_2 = 2 pF (fig.). Calculate the equivalent capacitance of the given network between points A and B.



Sol: Capacity C_1 = 3pF, Capacity C_2 = 2pF

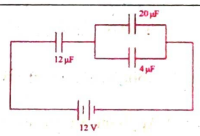
Initially C_1, C_2, C_1 are in series $\Rightarrow \frac{1}{C_s} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_1} = \frac{1}{3} + \frac{1}{2} + \frac{1}{3} = \frac{2+3+2}{6}$ $\therefore C_S = \frac{6}{7}pF$

Now C_S and C_2 are in parallel.

$\therefore C_P = C_s + C_2 = \frac{6}{7} + 2 = \frac{6+14}{7} = \frac{20}{7}pF$ Then C_1, C_P, C_1 are in series.

---

30. In the circuit shown in fig. the charge on the capacitor of 4 μF is 16 μC. Calculate the energy stored in the capacitor of 12 μF capacitance.



Sol: Given that, charge on the capacitor of 4μF is of 16μC.

Voltage across 4 μF capacitor. $V_2 = \frac{q}{C} = \frac{16}{4} = 4V$

$\therefore$ Voltage across 12 μF capacitor $V_1 = V - V_2 = 12 - 4 = 8V$

$\therefore$ Energy stored in the capacitor 12μF is $U = \frac{1}{2}CV_1^2 = \frac{1}{2}(12)(8)^2 = 384\,\mu J$

$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_P} + \frac{1}{C_1} = \frac{1}{3} + \frac{7}{20} + \frac{1}{3} = \frac{20+21+20}{60} = \frac{61}{60}$ $\therefore$ Resultant capacity $C = \frac{60}{61}pF$

---

31. The electric potential at 0.9m from a point charge is +50V. What is the magnitude and sign of the charge?

Sol: Given potential V = 50V, distance r = 0.9m

Potential $V = K\frac{q}{r} \Rightarrow q = \frac{Vr}{K} = \frac{50\times0.9}{9\times10^9} = 5\times10^{-9}C$

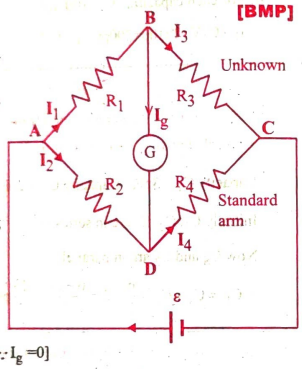
Since the given potential is positive, the charge must also be positive.

[☞ Remaining SAQs are in Page 99]

3 CURRENT ELECTRICITY

4 Marks

32. Using Kirchhoff's rules deduce the condition for balance in a Wheatstone bridge. **[BMP]**



A: **1) Wheatstone's Bridge** consists of four resistors R_1, R_2, R_3, R_4 with four junctions A,B,C, D as shown in the figure.

2) AC is battery arm, BD is galvanometer arm. When the bridge is balanced, the current flowing through the galvanometer $I_g = 0$,.

3) Applying Kirchhoff's first law at B,

we have $I_1 - I_g - I_3 = 0 \Rightarrow I_1 = I_3$ .....(1) [$\because I_g = 0$]

4) Applying Kirchhoff's first law at D, we have $I_2 + I_g - I_4 = 0 \Rightarrow I_2 = I_4$ .....(2) [$\because I_g = 0$]

5) Applying Kirchhoff's second law to closed loop ABDA, we have $I_1 R_1 + I_g G - I_2 R_2 = 0$

$\Rightarrow I_1 R_1 = I_2 R_2 \Rightarrow \frac{I_1}{I_2} = \frac{R_2}{R_1}$ .....(3)[$\because I_g = 0$]

6) Applying Kirchhoff's second law to closed loop BDCB, we have $I_g G + I_4 R_4 - I_3 R_3 = 0$

$\Rightarrow I_3 R_3 = I_4 R_4 = \Rightarrow \frac{I_3}{I_4} = \frac{R_4}{R_3}$ .....(4)

7) But from (1) & (2), $I_1 = I_3$ and $I_2 = I_4$ From (4), we have $\frac{I_1}{I_2} = \frac{R_4}{R_3}$ .....(5)

8) Now from (3) & (5) we get $\frac{R_2}{R_1} = \frac{R_4}{R_3}$ .....(6)

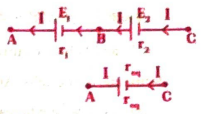
9) $\therefore \frac{R_1}{R_2} = \frac{R_3}{R4}$ (or) $\frac{R_1}{R_3} = \frac{R_2}{R4}$. This is the balance condition of Wheatstone's bridge to make $I_g = 0$.

[☞ Remaining SAQs are in Page 99]

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33. Derive equivalent emf and internal resistance of two cells in series combination.

A: 1) Consider two cells of emf's E_1 and E_2 and their internal resistances r_1 , r_2 connected in series. Let V_A , V_B , V_C be the potentials at points A, B, C .



2) The potential difference between positive and negative terminals of the first cells between A and B is

$V_{AB} = V_A - V_B = E_1 - I r_1$ .....(1)

Similarly, $V_{BC} = V_B - V_C = E_2 - I r_2$ .....(2)

3) The potential difference between positive and negative terminals of A and C is

$V_{AC} = V_A - V_C = (V_A - V_B) + (V_B - V_C) = (E_1 - Ir_1) + (E_2 - Ir_2)$

$\therefore V_{AC} = (E_1 + E_2) - I[r_1 + r_2]$.....(3)

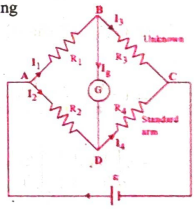
4) If we replace the combination by a single cell between A and C of emf (E_{eq}) and internal resistance (r_{eq}) then $V_{AC} = E_{eq} - Ir_{eq}$ .....(4)

5) Comparing (3) and (4), we get $E_{eq} = E_1 + E_2$ and $r_{eq} = r_1 + r_2$

---

34. State and explain Kirchhoff rules for an electric network.

A: **1) Kirchhoff's First Law :** The algebraic sum of currents meeting at any junction in an electric circuit is zero.



2) Applying **Kirchhoff's first law at B,**

we have $I_1 - I_g - I_3 = 0$

3) Kirchhoff's Second Law : The algebraic sum of changes in potential around any closed loop involving resistors and cells in the loop is zero.

4) Applying **Kirchhoff's second law to closed loop ABDA,**

we have $I_1 R_1 + I_g G - I_2 R_2 = 0$

---

35. When the resistance connected in series with the cell is halved, the current is equal to or slightly less or slightly greater than double, Why?

A: 1) The current in a circuit with a battery of emf E, internal resistance r and external resistance R is given by $I = \frac{E}{R + r}$

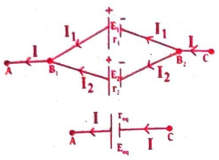
2) When the resistance is halved (R/2) we have the new current $I' = \frac{E}{(R/2) + r}$

3) If r is negligible when compared to R/2 then we get $I' = \frac{2E}{R} = 2I$ $(\because \frac{E}{R} = I)$

4) Also if r<<(R/2) then the new current I' is slightly greater than 2I and if r>R then the new current I' is slightly less than 2I.

36. Derive equivalent emf and internal resistance of two cells in parallel combination.

Sol: 1) Resistors in Parallel : Consider two cells of emf's E_1 and E_2 and internal resistance r_1 and r_2 connected in parallel as shown in fig.



2) Let I_1 and I_2 be the currents leaving the positive terminals of the cells. At junction B_1 the currents I_1 and I_2 flow in whereas the current I flows out.

Total current at junction B is $I = I_1 + I_2$ .....(1)

3) Let V_{B_1} and V_{B_2} be the potentials at B_1 and B_2.

For the first cell, potential difference is $V = V_{B_1} - V_{B_2} = E_1 - I_1 r_1$

$$\Rightarrow I_1 r_1 = E_1 - V \Rightarrow I_1 = \frac{E_1 - V}{r_1} \quad .....(2)$$

4) For the second cell, potential differences is $V = V_{B_1} - V_{B_2} = E_2 - I_2 r_2$

$$\Rightarrow I_2 r_2 = E_2 - V \Rightarrow I_2 = \frac{E_2 - V}{r_2} \quad .....(3)$$

5) From (1), (2) and (3) we have $I = I_1 + I_2 = \frac{E_1 - V}{r_1} + \frac{E_2 - V}{r_2}$

$$\Rightarrow I = \frac{E_1}{r_1} - \frac{V}{r_1} + \frac{E_2}{r_2} - \frac{V}{r_2} \Rightarrow I = \left[\frac{E_1}{r_1} + \frac{E_2}{r_2}\right] - V\left[\frac{1}{r_1} + \frac{1}{r_2}\right]$$

$$\Rightarrow V\left[\frac{1}{r_1} + \frac{1}{r_2}\right] = \left[\frac{E_1}{r_1} + \frac{E_2}{r_2}\right] - I$$

$$\Rightarrow V\left[\frac{r_1 + r_2}{r_1 r_2}\right] = \left[\frac{E_1 r_2 + E_2 r_1}{r_1 r_2}\right] - I = \left[\frac{(E_1 r_2 + E_2 r_1) - I r_1 r_2}{r_1 r_2}\right]$$

$$\Rightarrow V = \left[\frac{\cancel{r_1 r_2}}{r_1 + r_2}\right]\left[\frac{(E_1 r_2 + E_2 r_1) - I r_1 r_2}{\cancel{r_1 r_2}}\right] \Rightarrow V = \left[\frac{E_1 r_2 + E_2 r_1}{r_1 + r_2}\right] - I\left[\frac{r_1 r_2}{r_1 + r_2}\right] \quad .....(4)$$

6) If we replace the combination by a single cell between A and C of emf (E_{eq}) and internal resistance (r_{eq}) then $V_{AC} = E_{eq} - I r_{eq}$ .....(5)

7) Comparing (4) and (5) we get $E_{eq} = \frac{E_1 r_2 + E_2 r_1}{r_1 + r_2}$ and $r_{eq} = \frac{r_1 r_2}{r_1 + r_2}$

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4 MAGNETISM & MATTER

4 Marks

37. Mention the properties of magnetic field lines.

A: Properties of magnetic field lines :

1) The magnetic field lines of a magnet form continuous closed loops.
2) The magnetic line of force starts from a north pole and ends at south pole outside the magnet and will be from south pole to north pole inside the magnet.
3) At a given point of the field line, the tangent represents the direction of the field at that point.
4) The magnetic field lines don't intersect. If they intersect, magnetic field at that point will have two directions which is not possible.
5) The magnetic lines of force experience longitudinal strain. Due to this property unlike poles attract each other.
6) The magnetic lines of force exert lateral pressure. Due to this property like poles repel each other.

---

38. Explain the properties of diamagnetic materials.

A: Properties of diamagnetic substances:

1) Diamagnetic substances are weakly magnetised in opposite direction to that of magnetising field
   Ex: Cu,Ag,Au,H_2O
2) They are feebly repelled by a magnet.
3) They move from higher intensity to lower intensity in a non-uniform magnetic field.
4) They are independent of temperature.
5) Their relative permeability $\mu_r < 1$, ($\chi < 0$, χ is small and negative)
6) Resultant magnetic moment of its atom is zero, in the absence of external field.

---

39. Explain the properties of paramagnetic materials. **[BMP]**

A: Properties of Paramagnetic substances:

1) Paramagnetic substances are weakly magnetised in the direction of magnetising field.
   Ex: Al, Na, Pt.
2) They are feebly attracted by a magnet.
3) They move from lower intensity to higher intensity in a non-uniform magnetic field.
4) They are dependent on temperature.
5) Their relative permeability $\mu_r > 1$., ($\chi > 0$, χ is small and positive)
6) Resultant magnetic moment of its atom is non-zero, due to unpaired electrons.

40. How the existence of domains, explain the behavior of ferromagnetic material?

A: **Ferromagnetic materials** are the substances which are strongly attracted by magnets.
Their atomic dipoles align spontaneously to form **strong clusters called domains.**

1) In ferromagnetic materials the atoms form innumerable small effective regions called "domains".
2) The size of domain varies from about 10^{-6} m to 10^{-3}m.
3) Each domain contains 10^{17} to 10^{21} atomic magnets aligned in the same direction.
4) In an external magnetic field, the resultant magnetism increases due to reorientation of all domains into the direction of field.
5) In the absence of external field the domains are oriented randomly, so that net magnetic moment is zero.

---

41. State Gauss' law in magnetism. Show that how this law proved the non-existence of magnetic monopoles.

A: **1) Gauss' Law for Magnetism:** Gauss' Law for magnetism states that the net magnetic flux through any closed surface is always zero.
Gauss law for magnetism is $\oint \vec{B}.d\vec{S} = 0$

2) Explanation: The above equation states that the net magnetic flux through any closed surface is zero. Magnetic filed lines always form closed continues loops. They do not begin or end at any point, unlike electric field lines.

3) Non-existence of Magnetic Monopoles: Since the net flux through a closed surface is zero there is no isolated magnetic poles (monopoles) in nature. Magnetic poles always exist in pairs (North & south)

4) Conclusion: Gauss law in Magnetism proves that magnetic monopoles do not exist.

---

42. Define magnetisation of a sample, magnetic induction and susceptibility. How magnetic induction is related to magnetic intensity?

A: **1) Magnetisation (M)** of a sample is defined as the net magnetic moment per unit volume.

Formula: $M = \frac{m_{net}}{V}$ **SI unit:** A m^{-1}.

2) Magnetic induction(B) is defined as the total number of magnetic lines of force passing normally through a unit area of a substance. **SI unit:** Tesla (T)

3) Magnetic Susceptibility (χ) is defined as the ratio between magnetisation (M) and magnetic intensity (H). **Formula:** $\chi = \frac{M}{H}$. It has no units

4) Relation between (B) and (H):
In a material of a susceptibility χ_m, magnetic induction B is proportional to the magnetic intensity H
Thus $B \propto H \Rightarrow B = \mu H$; Here μ= permeability of the substance. In free space, $B = \mu_0 H$

5) The total magnetic induction B = B (applied field) + B (Magnetization)

$\therefore B = \mu_0 H + \mu_0 H \chi_m = \mu_0 H(1+\chi_m) = \mu_0(1+\chi_m)H = \mu_0(\mu_r)H$ [$\because 1+\chi_m = \mu_r$]

$\therefore B = \mu H$ [$\because \mu_0(\mu_r) = \mu$]

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5 ELECTRO MAGNETIC INDUCTION

4 Marks

43. Explain the Faradays law of induction and Lenz's law.

A: **1) Faraday's law of Electromagnetic Induction: The** magnitude of the induced emf in a circuit is equal to the time rate of change of magnetic flux through the circuit.

2) Mathematically, the induced emf is given by $\varepsilon = -\frac{d\phi_B}{dt}$

3) In the case of a coil of N turns, $\varepsilon = -N\frac{d\phi_B}{dt}$

4) Lenz's Law(Neumann's law) : The polarity of induced emf is such that it tends to produce a current which opposes the change in magnetic flux.

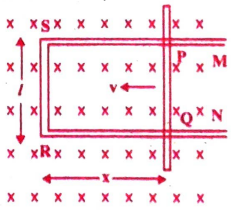
5) Faraday's law of induction is $\varepsilon = -\frac{d\phi_B}{dt}$. The negative sign in it indicates Lenz's law.

---

44. Derive an expression for motional emf induced across the conductor moving perpendicular to uniform magnetic field.

A: **Induced emf due to motion of a conductor in a magnetic field:**

1) Consider a rectangular conductor PQRS suspended in a uniform magnetic field B which is perpendicular to the plane of the system.



2) The conductor PQ is free to move. Suppose, the conductor PQ is moved towards left with velocity v.

3) Let RQ = x, RS = l.
Then the magnetic flux enclosed by the loop PQRS is $\phi_B = B\,l\,x$

4) The induced emf is given by $\varepsilon = -\frac{d\phi_B}{dt} = -\frac{d}{dt}(B\,l\,x) \Rightarrow \varepsilon = -B\,l\frac{dx}{dt}$. But $\frac{dx}{dt} = -v$

5) Thus the induced emf in a conductor moving in a perpendicular and uniform magnetic field is given by $\varepsilon = B\,l\,v$

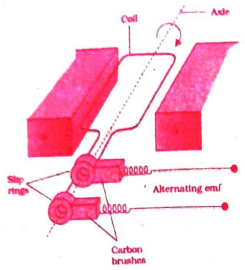
45. Explain briefly working of AC generator. [BMP]

A: AC Generator : AC generator converts **mechanical energy** into **electrical energy.**

Construction:

1) AC generator consists of a rectangular coil is placed between two magnetic poles.

2) The ends of the coil are connected to slip rings and brushes.



AC Generator

Working:

3) When the coil is rotated with a constant angular speed ω, θ = ωt.
where θ is the angle between vector area A and the magnetic field B at given time t.
The flux at time t is given by
$\phi_B = BA\cos\theta = BA\cos\omega t$ $(\because \theta = \omega t)$

4) From Faraday's law, the induced emf is $\varepsilon = -N\frac{d\phi_B}{dt} = -N\frac{d}{dt}(BA\cos\omega t)$
$= -NBA\frac{d}{dt}(\cos\omega t) = -NBA\ (-\sin\omega t)(\omega) = NBA\,\omega\sin\omega t.$

5) Thus the instantaneous emf is $\varepsilon = NBA\,\omega\sin\omega t$.
Here, $NBA\omega = \varepsilon_m$ = maximum emf.
$\therefore \varepsilon = \varepsilon_m \sin\omega t$ $(\because \varepsilon_m = NBA\omega)$
Hence the voltage produced by the generator is Sinusoidal.

---

46. Obtain an expression for the magnetic energy stored in a solenoid in terms of the magnetic field, area and length of the solenoid.

A : Magnetic Energy stored in a Solenoid:

1) The energy required to build up the current in a solenoid is given by $U = \frac{1}{2}LI^2$ .....(1)
where L is the self inductance of the solenoid.

2) Self inductance of a solenoid is given by $L = \mu_0 n^2 A l$ ..... (2)
here n is number of turns per unit length, A is area of cross section, l is length of the solenoid.
Substituting eq (2) in eq (1), we get $U = \frac{1}{2}(\mu_0 n^2 A l)I^2$ ..... (3)

3) Magnetic induction inside a solenoid is $B = \mu_0 n I \Rightarrow I = B/\mu_0 n$ ..... (4)

4) Putting this in eq (3), we get $U = \frac{1}{2}(\mu_0 n^2 A l)\frac{B^2}{\mu_0^2 n^2} = \frac{1}{2\mu_0}B^2 Al$

This is the expression for magnetic energy stored in a solenoid.

[☞ Remaining SAQs are in Page 100]

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6 ALTERNATING CURRENT

4 Marks

47. State the principle on which a transformer works. [BMP]
Describe the working of a transformer with necessary theory.

A : 1) Transformer: Transformer is a device which transforms AC electric power from one circuit to another. A transformer consists of two coils called primary coil and secondary coil wound on a continuous iron core.

2) Principle: Transformer is based on the principle of **mutual induction**. When current in primary coil changes, an induced current is produced in the secondary coil.

Transformer ratio $\frac{N_s}{N_p} = \frac{V_s}{V_p}$ Here N_s = number of turns in secondary coil,
N_p = number of turns in primary coil.
V_s = AC voltage in secondary, V_p = AC voltage applied to primary,

3) Working: When an alternating voltage V_p is applied to the primary, the resulting current produces an alternating magnetic flux which links the secondary and induces an emf ε_s in it. Let ϕ be the flux in each turn in the core at time t.

4) From Faraday's laws of induction, induced emf in the secondary coil is $\varepsilon_s = -N_s\frac{d\phi}{dt}$ .....(1)
Emf in the primary, $\varepsilon_p = -N_p\frac{d\phi}{dt}$ .....(2) (ϕ is alternating flux)

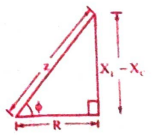
5) Here, $\varepsilon_s = V_s$ and $\varepsilon_p = V_p$ From (1) & (2) $V_s = -N_s\frac{d\phi}{dt}$ and $V_p = -N_p\frac{d\phi}{dt}$

6) Dividing them we get $\frac{V_s}{V_p} = \frac{N_s}{N_p}$. This is the formula for **transformer ratio.**

---

48. What is power factor? Derive its expression in terms of resistance R and impedance Z. Explain its significance in AC circuits.

A: 1) Power factor of an AC circuit is defined as the cosine of the phase angle (ϕ) between voltage and current. **Formula:** $\cos\phi = \frac{R}{Z}$



2) Expression: Consider impedance triangle for series LCR circuit as shown in the figure.
Here Z= hypotenuse, base =R,
perpendicular= $X_L - X_C$ and ϕ is the phase angle.
From the right angled triangle Power factor. $\cos\phi = \frac{R}{Z}$

3) Significance: If power factor $\cos\phi = 1$ the maximum power is dissipated in the circuit. If power factor $\cos\phi = 0$ then no power is consumed. This leads to the concept of wattless current dissipated even though current is flowing in the circuit

49. Obtain an expression for the current through an inductor when an AC emf is applied.

A : Current through an Inductor with AC emf :

1) Consider a pure inductor with self inductance L connected to an AC of voltage $v = v_m \sin\omega t$

2) Let I be the instantaneous current at any time t passing through the inductor.

Applying the Krichhoffs law on the circuit, we have $v - L\frac{dI}{dt} = 0$....(1)

$\Rightarrow L\frac{dI}{dt} = v \Rightarrow \frac{dI}{dt} = \frac{v}{L} = \frac{v_m \sin\omega t}{L}$ $(\because v = v_m \sin\omega t)$

3) Integrating it with respect to time, we get

$\int \frac{dI}{dt}dt = \frac{v_m}{L}\int \sin(\omega t)dt \Rightarrow I = \frac{v_m}{L}\left(\frac{-\cos(\omega t)}{\omega}\right)$ $\left(\because \int \sin kx = \frac{-\cos kx}{k}\right)$

$\Rightarrow I = \frac{v_m}{\omega L}\sin\left(\omega t - \frac{\pi}{2}\right)$....(2) $\left[\because -\cos(\omega t) = \sin\left(\omega t - \frac{\pi}{2}\right)\right]$

4) We know amplitude of current $I_m = \frac{v_m}{\omega L}$ $\therefore (2) \Rightarrow I = I_m \sin\left(\omega t - \frac{\pi}{2}\right)$

Current in an AC inductor circuit is $I = I_m \sin\left(\omega t - \frac{\pi}{2}\right)$

Comparing this with $v = v_m \sin\omega t$, we claim that the current **lags** the voltage by $\pi/2$.

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50. Obtain an expression for the current in a capacitor when an AC emf is applied.

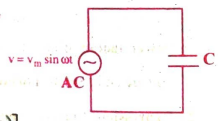
A : Current through an Capacitor with AC :

1) Consider a pure capacitor C connected to an AC voltage $v = v_m \sin\omega t$ .....(1)
Let I be the instantaneous current at any time t passing through the capacitor.

2) The potential difference across the capacitor of capacitance C is $v = \frac{q}{C}$ .....(2)

From (1) & (2), $\frac{q}{C} = v_m \sin\omega t$ $(\because v = v_m \sin\omega t) \Rightarrow q = C\ v_m \sin\omega t$

3) We know $I = \frac{dq}{dt} \Rightarrow I = \frac{d}{dt}(C\ v_m \sin\omega t)$



$\Rightarrow I = \omega C\, v_m \cos(\omega t)$ $\left(\because \frac{d}{dx}\sin kx = k\cos kx\right)$

$I = \omega C\, v_m \sin\left(\omega t + \frac{\pi}{2}\right)$....(2) $\left[\because \cos(\omega t) = \sin\left(\omega t + \frac{\pi}{2}\right)\right]$

4) We know amplitude of current $I_m = \frac{v_m}{X_C} = \frac{v_m}{1/\omega C} = v_m \omega C$

$(2) \Rightarrow I = I_m \sin\left(\omega t + \frac{\pi}{2}\right)$

5) Comparing this with $v = v_m \sin\omega t$, we claim that the current **leads** the voltage by $\pi/2$.

[☞ Remaining SAQs are in Page 100]

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7 WAVE OPTICS

4 Marks

51. Derive an expression for the intensity at a point where interference of light occurs. Arrive the conditions for maximum and zero intensity. [AP 16,18]

A : Expression for Intensity at a point due Interference of Light :

1) Consider two interfering light rays coming from two sources.
The displacements of the two rays are given by $y_1 = A\cos\omega t$, $y_2 = A\cos(\omega t + \phi)$
where a is amplitude, ω is angular frequency and ϕ is the phase difference between the waves.

2) From the principle of superposition of the rays the resultant displacement is $y = y_1 + y_2$
$\Rightarrow y = A[\cos\omega t + \cos(\omega t + \phi)]$

$\Rightarrow y = \left(2A\cos\left(\frac{\phi}{2}\right)\right)\cos\left(\omega t + \frac{\phi}{2}\right)$ $\left(\because \cos A + \cos B = 2\cos\left(\frac{A-B}{2}\right)\cos\left(\frac{A+B}{2}\right)\right)$

Here, $2A\cos\left(\frac{\phi}{2}\right)$ represents the amplitude of the resultant wave at the point.

3) The intensity of resultant wave is $I = 4I_0\cos^2\left(\frac{\phi}{2}\right)$ where I_0 is the intensity of each wave.

4) Conditions for Maximum Intensities:

The maximum value of resultant intensity I is $I_{max} = 4I_0$ when $\cos^2(\phi/2) = 1$.
Thus, the condition for maximum intensity at the point is $\phi = 0, \pm 2\pi, \pm 4\pi$ .....

5) Conditions for Zero Intensities:

The minimum value of resultant intensity I is $I_{min} = 0$ when $\cos^2(\phi/2) = 0$.
Thus, the condition for minimum intensity at the point is $\phi = \pm\pi, \pm 3\pi, \pm 5\pi$ . . . .

---

52. Does principle of conservation of energy hold for interference and diffraction phenomenon? Explain briefly. [BMP] [AP 16,17,20,25]

A : 1) Yes. The principle of conservation of energy holds good for both **interference** and **diffraction.**

2) In interference and diffraction, patterns of bright and dark fringes are formed. Here the principle of conservation of energy holds good for patterns of bright and dark fringes of interference and diffraction.

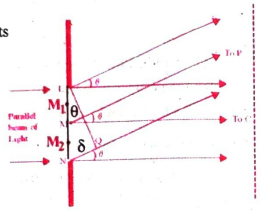
3) In interference and diffraction, light energy is redistributed. If it reduces in one region, producing a dark fringe, it increases in another region, producing bright fringe. Thus there is no gain or loss of energy. Hence, total energy remains constant.

53. Explain single slit diffraction pattern with neat diagram.
State conditions for maximum intensity and minimum intensity.

A: Diffraction at Single Slit:

1) Suppose a parallel beam of light is incident normally on a single slit LN of width a.
The diffracted light goes on to meet a screen at large distance.

2) Let us consider a point P on the screen which collects the waves originating from different points of the slit at an angle θ.



3) From the diagram,

$\sin\theta = \frac{\delta}{a} \Rightarrow \delta = a\sin\theta \Rightarrow \delta = a\theta$

(when θ is very small we have $\sin\theta = \theta$)

4) Condition for maximum intensity:

For maximum intensity $\delta = (2n+1)\frac{\lambda}{2}$ $\quad \therefore (2n+1)\frac{\lambda}{2} = a\theta \Rightarrow \theta = (2n+1)\frac{\lambda}{2a}$ $\quad ,n = 1,2,3,...$

5) Conditions for minimum intensity:

For minimum intensity $\delta = 2n\left(\frac{\lambda}{2}\right)$ $\quad \not{2}n\left(\frac{\lambda}{\not{2}}\right) = a\theta \Rightarrow \theta = \frac{n\lambda}{a}$, $\quad n=1,2,3,....$

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54. Discuss the intensity of light when a polaroid sheet is rotated between two crossed polaroids.

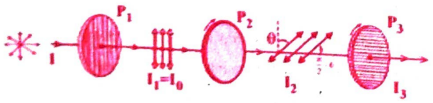
A: Intensity of Transmitted Polarised Light through a Polaroid :

1) Let I_0 be the intensity of polarised light after passing through the first polariser P_1.
Then the intensity of light after passing through second polariser P_2 will be $I_2 = I_0\cos^2\theta$
Here θ is the angle between pass axes of P_1 and P_2

2) Since P_1 and P_3 are crossed the angle between the pass axes of P_2 and P_3 will be $\left(\frac{\pi}{2}-\theta\right)$.
Hence the intensity of light emerging from P_3 will be $I_3 = I_2\cos^2(90°-\theta)$

$I_3 = (I_0\cos^2\theta)\sin^2\theta = \frac{I_0}{4}(4\sin^2\theta\cos^2\theta) = \frac{I_0}{4}\sin^2 2\theta$ $\quad [\because \sin 2\theta = 2\sin\theta\cos\theta]$

3) Intensity of transmitted light will be maximum when $\sin 2\theta = 1 \Rightarrow 2\theta = 90° \Rightarrow \theta = 45°$
∴ the transmitted intensity will be maximum when $\theta = \pi/4$



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8 DUAL NATURE OF RADIATION & MATTER

4 +4 Marks

55. What are different types of physical processes to obtain free electrons from the metal surface. **[BMP]**

A: Types of physical processes:

1) **Thermionic Emission:** By suitably heating, sufficient thermal energy can be imparted to the free electrons to enable them to come out of the metal.
Ex: Vacuum tubes, CRT & X rays tubes these are the applications of this emission.

2) **Field Emission:** By applying a very strong electric field to a metal, electrons can be pulled out of the metal, as in a spark plug.
Ex: Spark plugs, field emission microscopes, cold cathode devices are these applications.

3) **Photoelectric Emission:** When light of suitable frequency illuminates a metal surface, photo electrons are emitted from the metal surface.
Ex: Photoelectric cells, solar panels and light sensors are the applications of this emission.

4) **Secondary Emission:** When high speed 'primary' particles like electrons strike a metal surface they transfer their kinetic energy to the electrons within the metal, causing 'secondary' electrons to be knocked out of the surface.
Ex: Photo multiplier tubes, image intensifiers are the applications of this emission.

---

56. Define (i) Threshold frequency (ii) Work function and give the relation between them and define stopping potential.

A: i) **Threshold frequency:** It is the minimum frequency of incident radiation required to liberate electrons from the metal surface. It's value depends on the nature of the metal.

ii) **Work function:** It is the minimum amount of energy required just to liberate an electron from the given metal surface. Work function of a metal is a physical constant.

iii) **Relation between work function and threshold frequency:**
The work function (ϕ) is directly proportional to the threshold frequency (ν_0) of the metal.
Work function $\phi = h\nu_0$, h is planck's constant.

iv) **Stopping Potential:** It is the minimum negative potential given to the collector plate for which the photocurrent stops or becomes zero.

57. What are the laws of photoelectric effect.

A: Laws of photoelectric effect:

1) **Threshold Frequency:** For a given photosensitive surface, there exists a minimum frequency called threshold frequency (ν_0), below which photoelectric emission does not take place, whatever may be the intensity of incident light.

2) **Intensity :** The photoelectric current is directly proportional to the intensity of incident light, provided the frequency is greater than threshold frequency.

3) **Kinetic Energy :** The maximum kinetic energy of emitted photoelectrons depends on the frequency of incident radiation and is independent of the intensity of light. $K_{max}=h\nu-\phi$

4) **Instantaneous Emission:** The photoelectric emission is instantaneous. There is no measurable time lag between the incidence of light and the emission of photoelectrons.

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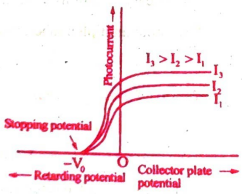
58. Explain graphically that the stopping potential is independent of intensity of incident light.

A: 1) Stopping Potential: It is the minimum negative potential given to the collector plate for which the photocurrent stops or becomes zero.

2) When light of the same frequency but different intensities falls on a photosensitive metal surface, the number of emitted photoelectrons changes then the corresponding photoelectric currents are measured.

3) Graphical Explanation:

i) We plot the photocurrent versus the collector potential (V) for light of same frequency and different intensities (I_1, I_2, I_3).



ii) From the graph it is clear that, as intensity of light increases ($I_3 > I_2 > I_1$), the saturation current also increases.

iii) The 3 curves converge at a negative potential point on the x-axis . This potential is called stopping potential V_0.

iv) From graph, the stopping potential is independent of intensity of incident light.



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59. Write Einstein's photo-electric equation.
How Millikan proved the validity of Einstein's photoelectric equation.

A: 1) Einstein's photoelectric equation: $h\nu=\phi+\frac{1}{2}mv^2_{max}$

Here h = Planck's constant, ν= frequency of incident radiation, f = work function of the metal

$\frac{1}{2}mv^2_{max}$ = maximum kinetic energy of emitted photoelectrons

2) Since maximum kinetic energy is related to stopping potential $V_0 \Rightarrow \frac{1}{2}mv^2_{max}=eV_0$

3) Einstein's equation can also be written as: $h\nu=\phi+eV_0 \Rightarrow V_0=\frac{h}{e}\nu-\frac{\phi}{e}$

4) Millikan's Verification of Einstein's Photoelectric Equation:

Millikan measured the stopping potential V_0 for different frequencies of incident light.

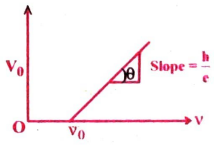
He plotted a graph between stopping potential V_0 and frequency ν

5) Observations:

i) The graph was a straight line.

ii) The slope of the graph was equal to h/e.

iii) The intercept on frequency axis gave the threshold frequency ν_0.



6) Conclusion:

The experimental results exactly matched the equation $V_0=\frac{h}{e}\nu-\frac{\phi}{e}$

Thus, Millikan proved the validity of Einstein's photoelectric equation.
He also determined the value of Planck's constant from the slope of the graph.

---

60. What is de Broglie's wavelength?

A: According to de Broglie's hypothesis, matter has dual nature . It states that every moving particle experiences a wave like nature.

2) de Broglie's wavelength $\lambda=\frac{h}{p}=\frac{h}{mv}$.

Here m is the mass of particle, v is the speed of particle, p is the momentum of particle.
h is the planck's constant, λ is the de Broglie wave length.
This equation is called de Broglie's relation.

3) For a microscopic particle like electron, the mass of the particle is extremely small.

4) Then the value of λ becomes very high as the mass m is in the denominator of the formula.
Hence, the moving particle behaves like a wave.

PROBLEMS

61. The work function of caesium is 2.14 eV. Find (a) the threshold frequency for caesium, and (b) the wavelength of the incident light if the photocurrent is brought to zero by a stopping potential of 0.60 V. [AP 16,24]

Sol: Given Work function $\phi_0 = 2.14\ eV = 2.14 \times 1.6\times10^{-19}J$, Stopping potential $V_0 = 0.60\ V$

Planck's constant $h = 6.63\times10^{-34}$ Js.

a) $\phi_0 = h\nu_0 \Rightarrow$ Threshold frequency $\nu_0 = \phi_0 / h = \dfrac{2.14\times1.6\times10^{-19}}{6.63\times10^{-34}} = 5.16\times10^{14}$ Hz

b) Einstein's photoelectric equation $eV_0 = h\nu - \phi_0 \Rightarrow h\nu = eV_0 + \phi_0 \Rightarrow \dfrac{hc}{\lambda} = eV_0 + \phi_0$

$\Rightarrow \lambda = \dfrac{hc}{(eV_0+\phi_0)} = \dfrac{(6.63\times10^{-34})(3\times10^{8})}{(0.60+2.14)1.6\times10^{-19}} = 4.54\times10^{-7}m = 454$ nm

---

62. The work function of a metal is 4.3 eV. What is the threshold wavelength?

Sol: Given work function $\phi = 4.3\ eV = 4.3\times1.6\times10^{-19} = 6.88\times10^{-19}J$

Planck's constant $h = 6.6\times10^{-34}$, Speed of light $c = 3\times10^{8}$ m/s

$\therefore$ Threshold wavelength $\lambda_0 = \dfrac{hc}{\phi} = \dfrac{12400}{4.3} = 2883.7\text{Å} = 288.37\times10^{-9}m = 288.37$nm.

---

63. A particle has a momentum of 1.5×10^{-27} kg m/s. What is its de-Broglie wavelength?

Sol: Given momentum of the particle $p = 1.5\times10^{-27}$ kg m/s, Planck's constant $h = 6.626\times10^{-34}$

$\therefore$ de-Broglie wave length $\lambda = \dfrac{h}{p} = \dfrac{6.626\times10^{-34}}{1.5\times10^{-27}} = 4.4173\times10^{-7} = 441.73\times10^{-9}$ m $= 441.73$ nm

---

64. What is the kinetic energy of an electron whose de-Broglie wavelength is 5000Å ? [BMP]

Sol: Given wavelength $\lambda = 5000\text{Å} = 5\times10^{-7}$m; Mass of electron $= 9.1\times10^{-31}$kg; $h = 6.63\times10^{-34}$ J/s

Kinetic energy $K = \dfrac{h^2}{2m\lambda^2}$ $\left(\because \lambda = \dfrac{h}{\sqrt{2mK}}\right)$

$\therefore K = \dfrac{[6.63\times10^{-34}]^2}{2\times[9.1\times10^{-31}][5\times10^{-7}]^2} = \dfrac{43.9569\times10^{-68}}{18.2\times10^{-31}\times25\times10^{-14}}$

$= \dfrac{43.9569}{455}\times10^{-23} = 0.0966\times10^{-23} = 9.66\times10^{-25}J$

$\therefore$ Kinetic energy K in eV $= \dfrac{9.66\times10^{-25}}{1.6\times10^{-19}}eV = 6.03\times10^{-6}$ eV

[☞ Remaining SAQ are in Page 102]

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9 ATOMS

4 Marks

65. State Bohr's postulates of hydrogen atom. [BMP]

A: Postulates of Bohr's Theory :

1) **First postulate:** Electron revolves around the positively charged nucleus in circular orbits called stationary orbits. Electrons do not radiate energy while revolving in stationary orbits.

For electron to revolve, Centripetal force= Electrostatic force of attraction

$\therefore \dfrac{mv^2}{r} = \dfrac{1}{4\pi\varepsilon_0}\dfrac{e^2}{r^2}$

2) **Second postulate:** Angular momentum of revolving electron around the nucleus is an integral multiple of $\dfrac{h}{2\pi}$.

$\therefore mvr = \dfrac{nh}{2\pi}$, $n = 1,2,3....$ represents the principal quantum number

3) **Third postulate:** When electron jumps from higher energy level to lower energy level, the energy will be radiated in the form a light photon

$\Delta E = E_2 - E_1 = h\nu$

Here E_1 and E_2 are the energies of the initial and the final states,

ν is the frequency of emitted radiation and h is Planck's constant.

---

66. What are the limitations of Bohr's theory of hydrogen atom? [AP 17,22,24]

A : Limitations of Bohr's theory of Hydrogen atom :

1) It could not explain the **fine structure of spectral lines** in hydrogen atom.
2) It could not explain the **elliptical orbits** because Bohr assumed the circular orbits.
3) It could not explain the **wave properties** of electron.
4) It could not explain the **splitting of spectra** under **electric and magnetic field.**

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67. What are the observations of Rutherford's α - particle scattering experiment and how these findings supported the existence of atomic nucleus.

A: **1) Observations of Rutherford's α - particle scattering :**

i) The majority of the α-particles passed directly through the gold foil without any deflection.

ii) A small fraction of α-particles were deflected by very small angles. (just more than 1°)

iii) Few α-particles were deflected by large angles (more than 90°) and very few (about 1 in 8000) bounced back.

2) Conclusions:

i) Most alpha particles passed straight through the gold foil. It shows that most of the space inside an atom is empty.

ii) Some positively charged alpha particles were deflected by small angles. It shows that the positive charge is concentrated at the centre of the atom.

iii) A tiny particle bounced straight back shows that it must have hit something extremely heavy and dense for which Rutherford called the nucleus of the atom.

By analyzing these findings, Rutherford concluded that an atom consists of a tiny, dense, positively charged the nucleus.

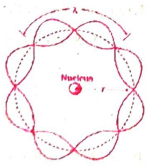
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68. Write a short note on debroglie's explanation of Bohr's second postulate of quantization?

A: **1) Bohr's second postulate:** The angular momentum of electron in a stationary orbit is quantised.

Thus, $m v_n r_n = \frac{nh}{2\pi}$, n=1,2,3, ...(1)

2) de -Broglie's explanation:



de- Broglie explained that the electron in the stationary orbit acts like a particle wave.

So it forms stationary waves in the orbit.

3) The distance travelled by the particle wave along circumference is an integral multiple of wavelength.

Thus, $2\pi r_n = n\lambda$ ....(2)

4) From de Broglie hypothesis, we have $\lambda = \frac{h}{p} = \frac{h}{mv_n}$

From (2), $2\pi r_n = n\left(\frac{h}{mv_n}\right) \Rightarrow m v_n r_n = \frac{nh}{2\pi}$ .....(3)

Thus, from (1) & (3) we conclude that de Broglie's explanation coincides with the second postulate of Bohr model of atom.

69. Derive expressions for potential and kinetic energies of an electron in any orbit of a hydrogen atom according to Bohr's atomic model.

A : 1) Consider an electron of H-atom of mass m, charge e.

Let it revolve in n^{th} orbit of radius r with a velocity v.

2) For electron to revolve, Centripetal force = Electrostatic force of attraction

$$\Rightarrow \frac{mv^2}{r} = \frac{1}{4\pi\varepsilon_0}\frac{e^2}{r^2} \Rightarrow mv^2 = \frac{1}{4\pi\varepsilon_0}\frac{e^2}{r} \ldots(1)$$

3) From Bohr's 2nd postulate, we have $mvr = \frac{nh}{2\pi} \Rightarrow m^2v^2r^2 = \frac{n^2h^2}{4\pi^2} \ldots(2)$

$$\frac{(2)}{(1)} \Rightarrow \frac{m^2v^2r^2}{mv^2} = \frac{n^2h^2 4\pi\varepsilon_0 r}{4\pi^2.e^2} \Rightarrow r = \frac{h^2\varepsilon_0 n^2}{\pi e^2 m} \quad .....(3)$$

a) K.E of electron: We know $K.E = \frac{1}{2}mv^2$

From (1), $K = \frac{1}{2}\left(\frac{1}{4\pi\varepsilon_0}\frac{e^2}{r}\right) = \frac{1}{8\pi\varepsilon_0}\frac{e^2}{r}$ .....(4)

Substituting value of r from eq(3) in (4) we get

Kinetic Energy $K = \frac{1}{8\pi\varepsilon_0}\frac{e^2.\pi e^2 m}{h^2\varepsilon_0 n^2} = \frac{me^4}{8\varepsilon_0^2 h^2 n^2}$

b) PE of electron: P.E=Potential (V) × charge of electron.

Potential energy $U = \frac{1}{4\pi\varepsilon_0}\frac{e}{r}(-e) = -\frac{1}{4\pi\varepsilon_0}\frac{e^2}{r}$ .....(5)

Substituting the value of radius r from (3) in eqn (5) we get

$$U = -\frac{e^2}{4\pi\varepsilon_0}\left(\frac{\pi e^2 m}{h^2\varepsilon_0 n^2}\right) = -\frac{me^4}{4\varepsilon_0^2 h^2 n^2}$$



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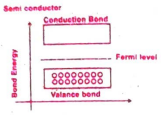
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10 SEMI CONDUCTOR ELECTRONICS

4 Marks

70. Describe energy band diagram of intrinsic semiconductors at 0 K.

A: Intrinsic semiconductors:



1) Intrinsic semiconductors are made up of pure form of semi conductor. These atoms are tetravalent.

They behave like insulators at 0K.

2) At 0K, the valence band is completely filled with electrons and conduction band is empty.

Thus there are no free electrons to conduct electricity. Hence they act as insulators at 0K.

3) At 0K there is a small energy gap between the top of the valence band and the bottom of the conduction band. Since, no thermal energy is available at 0 K, no electrons can jump across this gap.

4) At higher temperatures (T > 0K) thermal energy excites some electrons from valence band to the conduction band and leaving equal number of holes there.

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71. What are n-type and p-type semiconductors?

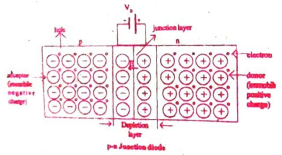
Explain the process of formation of p-n junction diode.

A: **1) n-type semiconductors** are silicon or germanium doped with pentavalent impurity like arsenic.

2) p-type semiconductors are silicon or germanium doped with trivalent impurity like indium.

3) Process of formation of p-n junction diode:

When a semiconductor material such as silicon or Germanium crystal is doped such that one side of it is p-type and the other side is n-type then the interface is called p-n junction and the device is called p-n junction diode.



4) When p-n junction is formed, the free electrons which are in higher concentration on n-side diffuse over to p-side and combine with holes and become neutral. Similarly, hole's which are in higher concentration on p-side diffuse over to n-side and combine with electrons and become neutral.

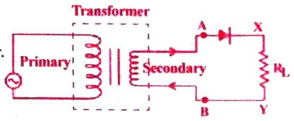
5) This results in the formation of a narrow region on either side of the junction which is free from mobile charge carriers. This region is called **depletion layer.** This layer acts as a potential barrier between p-type and n-type material of diode.

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72. Describe how a semiconductor diode is used as a half wave rectifier.

A: **1) Rectification:** The process of converting alternating current (AC) into direct current (DC) is called rectification.

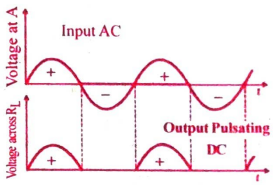
2) Half Wave Rectifier: A circuit which rectifies half of the AC wave is called Half wave rectifier.



3) Diode as a Halfwave Rectifier: A single diode is connected in series to a load resistance R_L in the secondary circuit of a transformer.

4) Working:

- During the positive half cycle, the diode is forward biased and the current flows through the diode.
- During the negative half cycle, the diode is reverse biased and the current does not flow through the diode.
- Thus, half wave is rectified using a single diode.



5) Efficiency of Halfwave rectifier $\eta = \frac{0.406R_L}{r_f + R_L}$

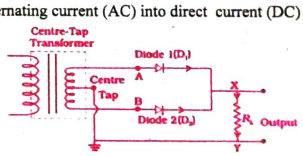
6) Maximum Efficiency of Halfwave rectifier is 40.6%

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73. What is rectification? Explain the working of a full wave rectifier.

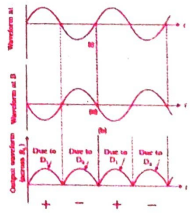
A: **1) Rectification:** The process of converting alternating current (AC) into direct current (DC) is called rectification.

2) Full Wave Rectifier: The circuit which rectifies both half cycles of an AC wave is called Full Wave Rectifier.



3) Working:

- During the positive half cycle, diode D_1 is forward biased & D_2 is reverse biased. Hence current passes through R_L.
- During the negative half cycle, the D_2 is forward biased and D_1 is reverse biased. Hence, current passes through R_L.
- Thus the full wave is rectified using two diodes.



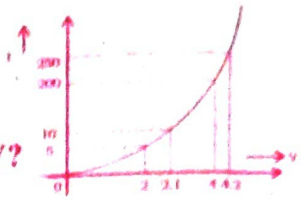
4) Efficiency of full wave rectifier is $\eta = \frac{0.812R_L}{r_f + R_L}$

Here r_f = diode forward resistance, R_L = load resistance.

5) Maximum Efficiency of full-wave rectifier is 81.2%

PROBLEMS

74. The I-V characteristics of a p-n junction diode in forward bias are shown in the following figure. Find the ratio of dynamic resistance corresponding to forward bias voltages of 2V and 4V?



[BMP]

A: (i) Dynamic resistance at 2V voltage

Voltage at $V_1 = 2V$; current $I_1 = 5mA$; Voltage at $V_2 = 2.1V$; current $I_2 = 10mA$

$\therefore \Delta V = V_2 - V_1 = 2.1 - 2 = 0.1\ V$

$\Delta I = I_2 - I_1 = 10 - 5 = 5 \times 10^{-3} A$ Resistance $Rd_1 = \frac{\Delta V}{\Delta I} = \frac{0.1}{5\times10^{-3}} = 20\Omega$

(ii) Dynamic resistance at 4V voltage.

Voltage at $V_3 = 4V$, current $I_3 = 200mA$; Voltage at $V_4 = 4.2V$, current $I_4 = 250mA$

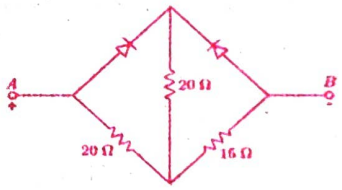
$\Delta V = V_4 - V_3 = 4.2 - 4 = 0.2V$

$\Delta I = I_4 - I_3 = 250 - 200 = 50 = 50 \times 10^{-3} A$

Resistance $Rd_2 = \frac{\Delta V}{\Delta I} = \frac{0.2}{50\times10^{-3}} = \frac{200}{50} = 4\Omega$ $\therefore$ Ratio $= \frac{Rd_1}{Rd_2} = \frac{20}{4} = 5:1$

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75. Two ideal diodes are connected in the network as shown in the figure. Then find the equivalent resistance between A and B?



A: 20Ω resistors are in parallel,

$\therefore \frac{1}{R_P} = \frac{1}{20} + \frac{1}{20} = \frac{2}{20} = \frac{1}{10} \Rightarrow R_P = 10\Omega$

Equivalent resistance $R_{eq} = R_P + 15\Omega = 10 + 15 = 25\Omega$

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76. Pure silicon at 300K has an equal number of electrons (n_i) and the hole (n_h) concentrations $5 \times 10^{16}\ m^{-3}$. When it was doped with the indium then the hole concentration (n_h) increases to $4.5 \times 10^{22}\ m^{-3}$ then find the concentration of electrons after doping.

A: Given $n_i = 5\times10^{16} m^{-3}$; $n_h = 4.5\times10^{22} m^{-3}$ $\therefore n_e = \frac{(n_i)^2}{n_h} = \frac{(5\times10^{16})^2}{4.5\times10^{22}} = 5.55\times10^{10} m^{-3}$

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77. A potential barrier of 0.8V exists across the p-n junction. If the depletion region is 800 nm wide, then what is the intensity of electric field with in this region is?

A: Barrier potential $V = 0.8V$

Width of depletion region $d = 800\ nm = 800 \times 10^{-9}\ m = 8 \times 10^{-7}\ m$

$\therefore$ Electric field Intensity $E = \frac{V}{d} = \frac{0.8}{8\times10^{-7}} = 10^6\ V/m$

[☞ Remaining SAQs are in Page 103]